

The Boris Pusher, and ODEs across Physics



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Today

- **The Boris pusher.** The standard integrator for charged particles, and why it conserves energy.
- **Orbits.** How NASA chose an integrator for spacecraft, and why it still uses it.
- **Molecular dynamics.** Velocity Verlet on a gas of atoms, and three things it shows.
- **Cosmology.** Why the microwave background is an ODE problem.

Part I · The Boris pusher



Charged particles

A charged particle in a magnetic field



$$\frac{d\mathbf{x}}{dt} = \mathbf{v}, \quad \frac{d\mathbf{v}}{dt} = \frac{q}{m}(\mathbf{E} + \mathbf{v} \times \mathbf{B})$$

- With $\mathbf{E} = 0$ the magnetic force is always perpendicular to $\mathbf{v}$: it does no work, and $|\mathbf{v}|$ never changes.
- In a uniform $\mathbf{B}$ the particle circles at the cyclotron frequency $\omega_c = qB/m$.
- A good method should keep $|\mathbf{v}|$ exactly, over millions of gyrations.

Leapfrog for the Lorentz force



Stagger as in Lecture 10: positions at whole steps, velocities at half steps.

$$\mathbf{x}_{n+1} = \mathbf{x}_n + \Delta t \mathbf{v}_{n+1/2}$$
$$\frac{\mathbf{v}_{n+1/2} - \mathbf{v}_{n-1/2}}{\Delta t} = \frac{q}{m} \left(\mathbf{E}_n + \frac{\mathbf{v}_{n+1/2} + \mathbf{v}_{n-1/2}}{2} \times \mathbf{B}_n \right)$$

- The force depends on the velocity, which lives at half steps, so it is averaged over the step.
- The unknown $\mathbf{v}_{n+1/2}$ appears on both sides: the update looks implicit.

Boris's trick (1970)



Half an electric kick, a rotation about $\mathbf{B}$, and another half kick:

$$\mathbf{v}^- = \mathbf{v}_{n-1/2} + \frac{q\mathbf{E}}{m} \frac{\Delta t}{2}$$

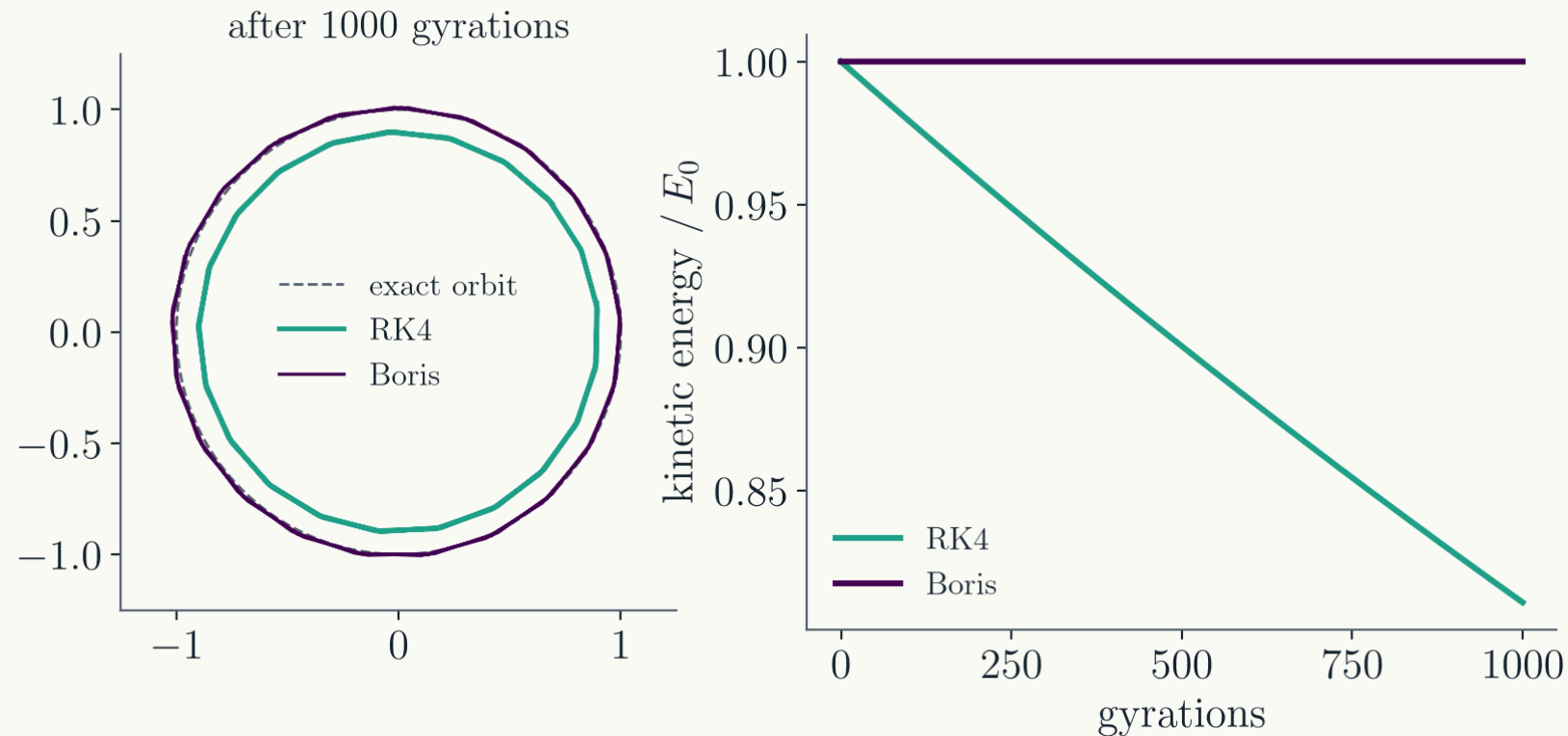
$$\mathbf{t} = \frac{q\mathbf{B}}{m} \frac{\Delta t}{2}, \quad \mathbf{s} = \frac{2\mathbf{t}}{1 + t^2}$$

$$\mathbf{v}^+ = \mathbf{v}^- + (\mathbf{v}^- + \mathbf{v}^- \times \mathbf{t}) \times \mathbf{s}$$

$$\mathbf{v}_{n+1/2} = \mathbf{v}^+ + \frac{q\mathbf{E}}{m} \frac{\Delta t}{2}$$

- Two cross products solve the "implicit" equation exactly: no iteration, no matrix.
- The middle step is an exact rotation, so $|\mathbf{v}^+| = |\mathbf{v}^-|$: with $\mathbf{E} = 0$ the energy is conserved to rounding.

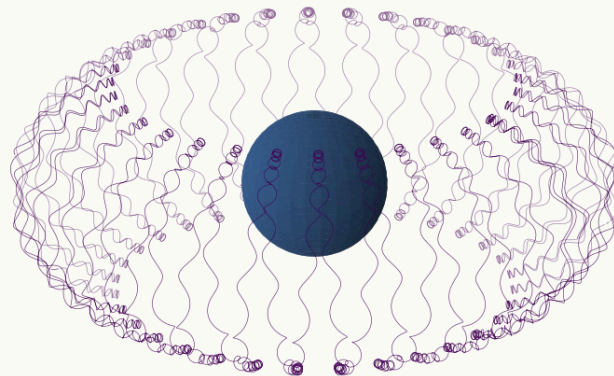
Boris against RK4, uniform field



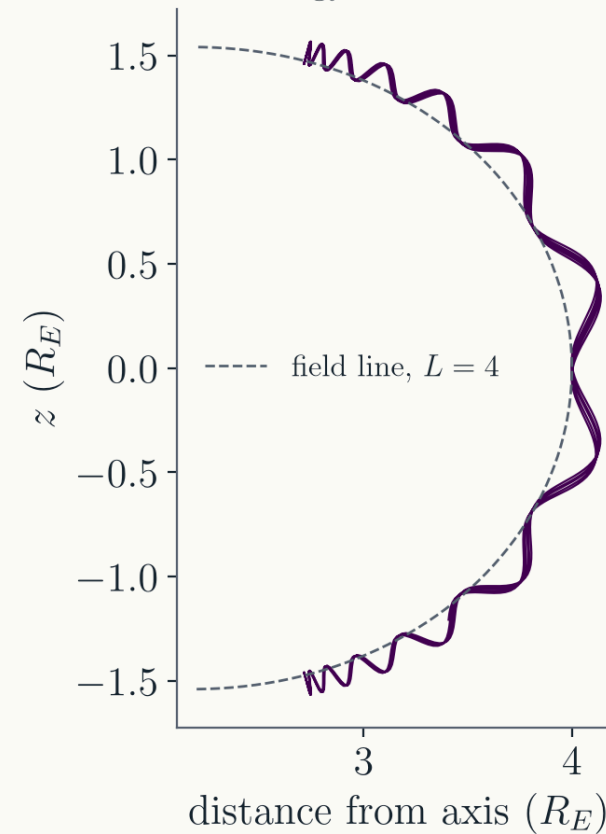
- $\omega_c \Delta t = 0.3$, about 21 steps per gyration, 1000 gyrations.
- RK4 loses a factor $1 - (\omega_c \Delta t)^6 / 72$ of energy per step (Lecture 10): 19% by the end. Boris: $|\mathbf{v}|$ exact to 10^{-13} .

A proton in the Van Allen belt

75 s: one drift around the Earth



first 6 s: gyration and bounce



10 MeV proton, $4 R_E$ from Earth's center

gyration

bounce

drift around the Earth

period

0.14 s

2.3 s

1.2 min

Three timescales, one integrator



- The step is set by the fastest motion, the gyration. The physics of interest (the drift that forms the belt) is 500 times slower.
- Over the drift, Boris conserves the energy to 10^{-14} . RK4, with the same step, loses 98% of it.
- This is why plasma codes push every particle with Boris.

Structure preservation matters most when the interesting timescale is far longer than the step.

Particle-in-cell simulations



Relativistic kinetic turbulence simulations ($\sigma=10$): density

Joonas Nättiä



Watch on

Relativistic plasma turbulence (J. Nättiä): youtube.com/watch?v=Gj9mwAww3TM

Billions of particles pushed by Boris (ODEs), and the fields on a grid, evolved by Maxwell's equations (PDEs): coming later in the course.

Research from our group



Space for new research results (1)

Research from our group



Space for new research results (2)

Part II · How NASA integrates orbits



Choosing an integrator in 1963

NASA's comparison, 1963

TABLE 1
Comparison Criteria

Method of Numerical Integration	Truncation Error	Ease of Changing Step-Size	Speed	Stability	Roundoff Error Accumulation
Single Step Methods					
Runge-Kutta	h^5	*	Slow	Stable	Satisfactory
Runge-Kutta Gill	h^5	*	Slow	Stable	Satisfactory
Bowie	h^3	Trivial (step-size varied by error control)	Fast	Stable	Satisfactory
Fourth Order Multistep Predictor-corrector					
Milne	h^5	Excellent	Very fast	Unstable	Poor
Adams-Moulton	h^5	Excellent	Very fast	Unconditionally stable	Satisfactory
Higher Order Multistep					
Adams Backward Difference	Arbitrary	Good	Very fast	Moderately stable	Satisfactory
Gauss-Jackson**	Arbitrary	Awkward and expensive	Fast	Stable	Excellent
Obrechhoff	h^7	Excellent	***	Stable	Satisfactory
Special Second Order Equations [$y'' = f(t, y)$]					
Special Runge-Kutta	h^5	*	Slow	Stable	Satisfactory
Milne-Stormer	h^6	Excellent	Very fast	Moderately stable	Poor

*R-K (single step) trivial to change steps, very difficult to determine proper size.
 **Gauss-Jackson is for second order equations.
 ***Speed of Obrechhoff depends on complexity of the higher order derivatives required; it could be very fast.

NASA Space Flight Handbook, vol. 1 (1963), Chapter IV.C: the methods of the last five lectures, rated for computing trajectories.

Runge-Kutta: easy to start, slow. Multistep: fast.

"Unconditionally stable" here means no growing parasitic roots at small h , not the stiff stability of Lecture 10.

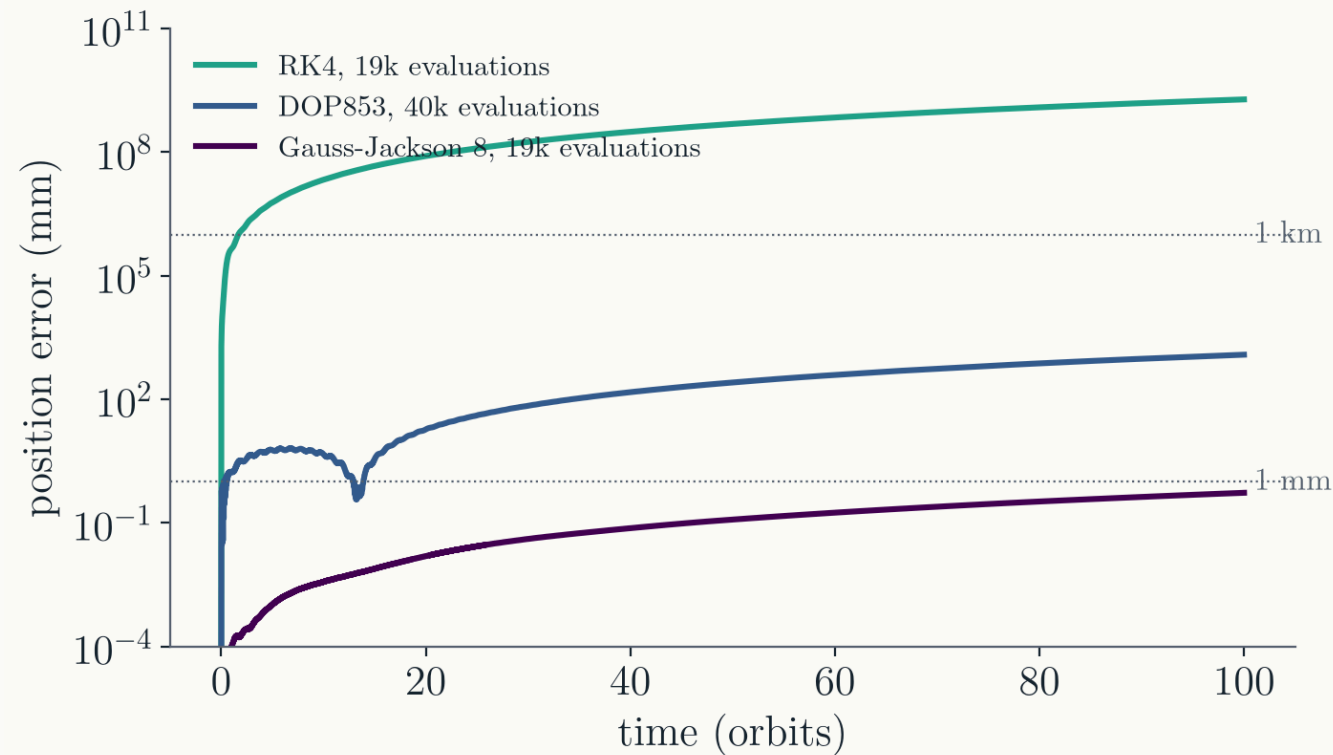
Their pick for orbits: Gauss-Jackson.

Gauss-Jackson



- A multistep predictor-corrector method (Lecture 10), built for second-order equations $\ddot{\mathbf{r}} = \mathbf{f}(t, \mathbf{r})$: it integrates twice without converting to a first-order system.
- Eighth order, fixed step, and about one or two evaluations of the force per step.
- Needs a start-up (a Runge-Kutta run for the first points), like every multistep method.
- Still used for satellite orbit propagation (Berry & Healy 2004).

100 orbits, measured



A circular orbit 7000 km from Earth's center (period 97 minutes), error after 100 orbits:

RK4, 20k evaluations	1900 km
DOP853, 40k	1.2 m
DOP853, 97k	0.7 mm
Gauss-Jackson 8, 20k	0.5 mm

High order, one or two evaluations per step, and a smooth periodic problem: the best case for a fixed-step multistep method.

Part III · Molecular dynamics



Newton's equations for many atoms

Molecular dynamics

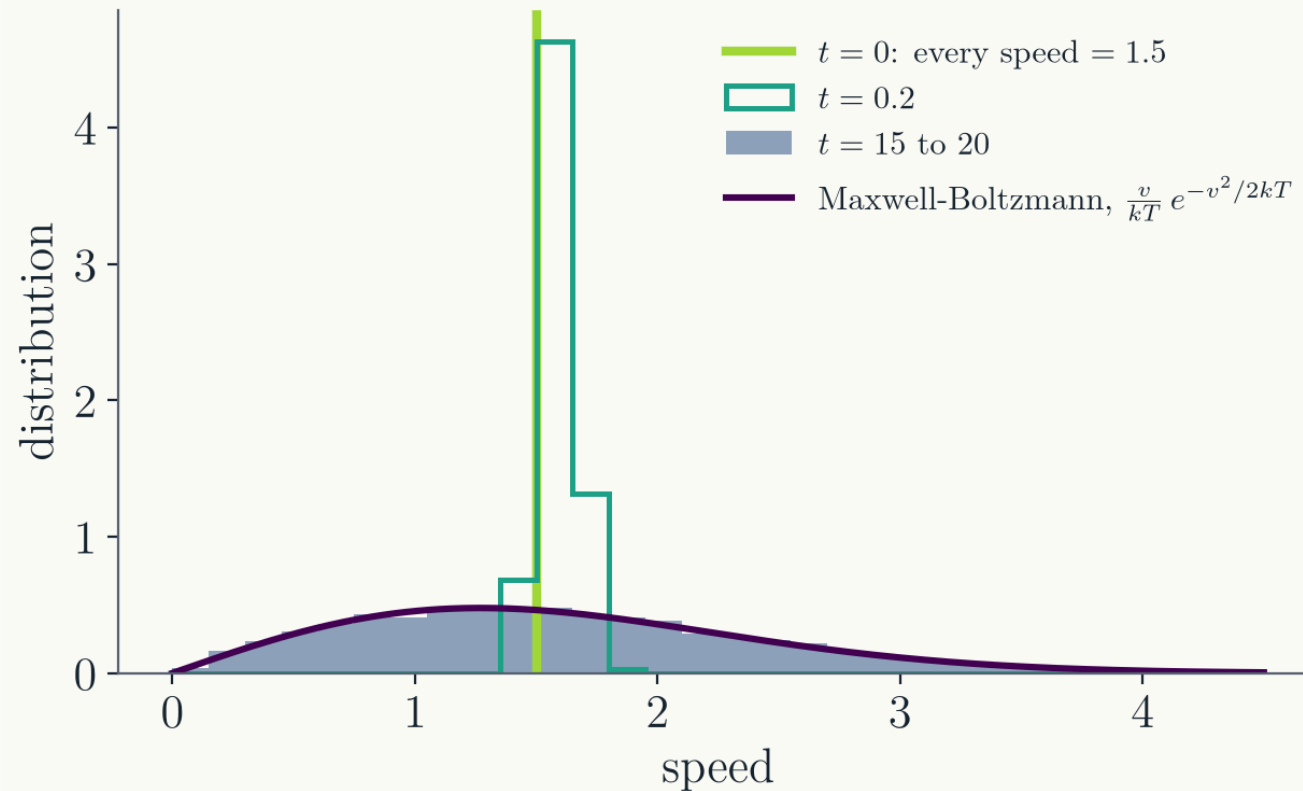


N atoms, Newton's second law, and a pair potential. For noble-gas atoms, Lennard-Jones:

$$m \ddot{\mathbf{r}}_i = - \sum_{j \neq i} \nabla_i V(r_{ij}), \quad V(r) = 4\epsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right]$$

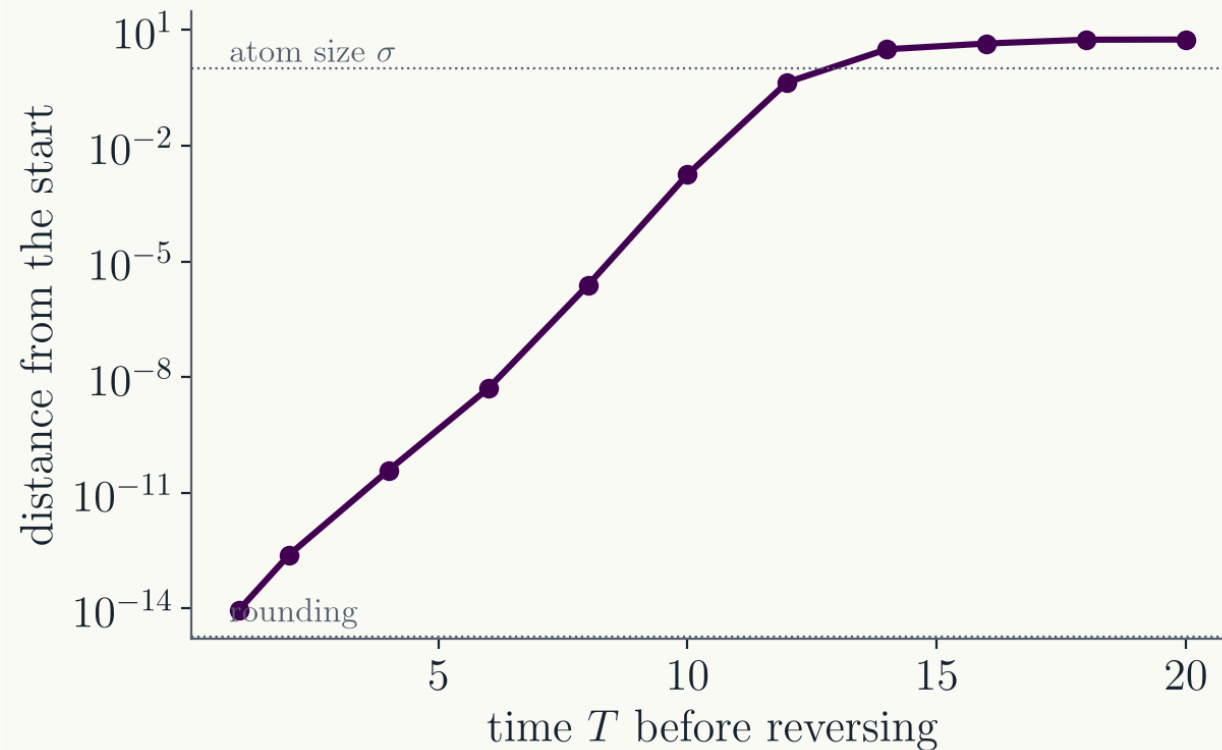
- Integrated with velocity Verlet (Lecture 10): one force evaluation per step, and bounded energy error over millions of steps.
- The cost is in the forces: all pairs is $O(N^2)$. Cutting the potential off at a few σ and keeping neighbor lists makes it $O(N)$.
- Used for liquids, materials, proteins and drug binding.

Collisions make Maxwell-Boltzmann



- 400 atoms, every one starting at the same speed, in random directions.
- Nothing statistical is put in: Newton's equations and collisions produce the Maxwell-Boltzmann distribution, at the temperature set by the energy.

Running the gas backwards

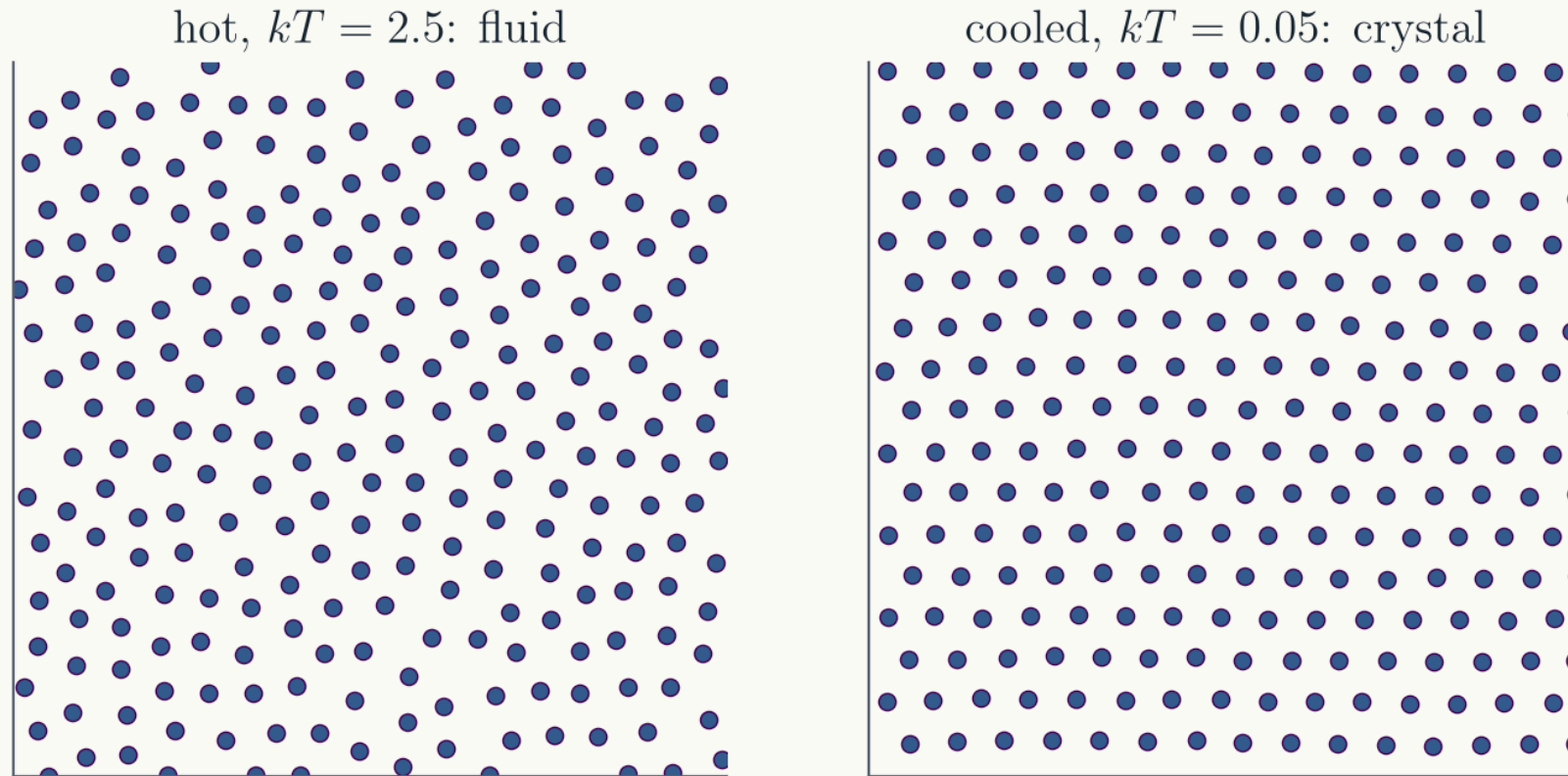


Run 100 atoms for a time T , reverse every velocity, run T again. Leapfrog is exactly reversible (Lecture 10), so the gas should return to where it started.

At $T = 1$ it does, to 10^{-14} . But every 0.35 time units the miss grows by a factor e : rounding errors are amplified by the chaos of the collisions.

By $T \approx 12$ the gas misses by half an atom, and after that it keeps no memory of its start.

Cooling into a crystal



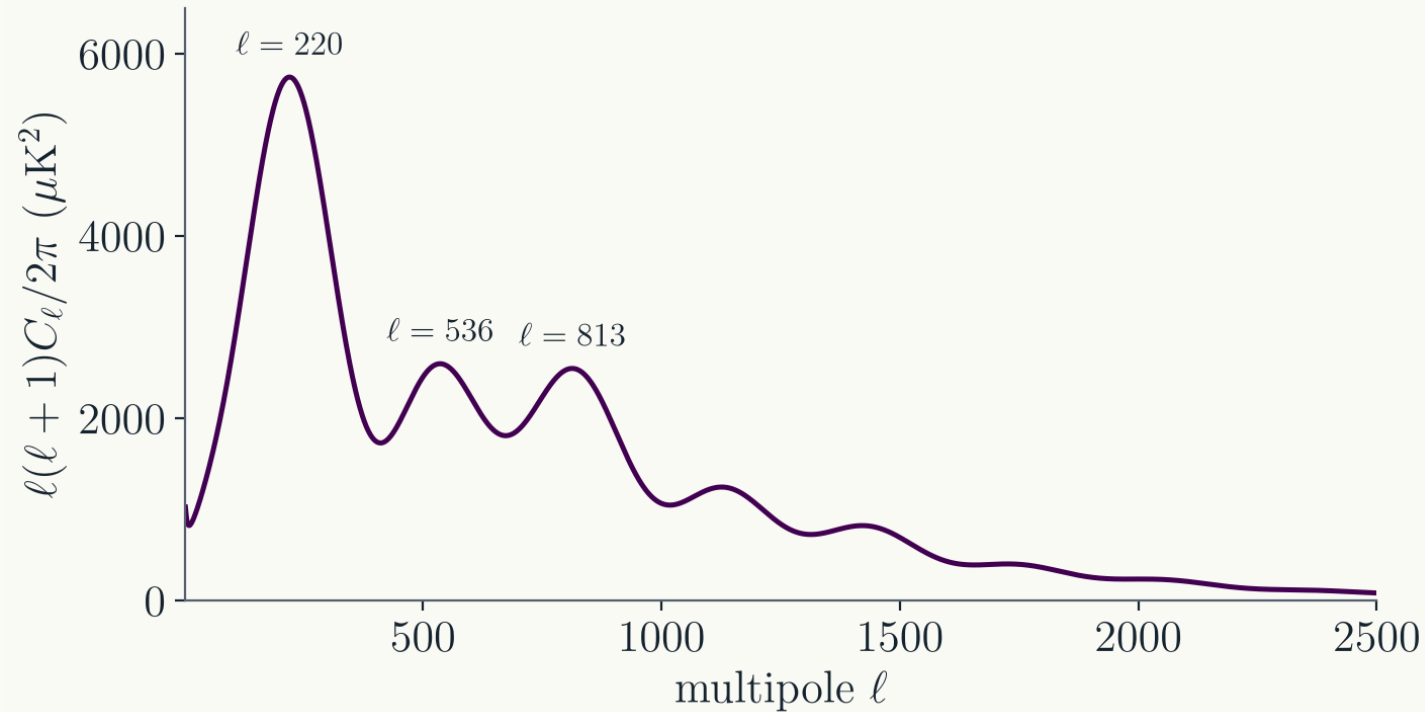
- 256 atoms, slowly cooled by rescaling their velocities every 50 steps.
- The atoms find the triangular lattice by themselves: freezing is in Newton's equations too.

Part IV · Cosmology



The universe as an ODE problem

The microwave background



- Temperature fluctuations of about 10^{-5} , left from when the universe became transparent, 380,000 years after the big bang.
- The peaks are sound waves in the early plasma. Their positions and heights measure the contents of the universe.

Why it is an ODE problem



- On large scales the universe is homogeneous: the background (expansion, density, temperature) depends on time alone.
- The fluctuations are small, so their equations are linear. Fourier transform in space: each wavenumber k evolves on its own.
- For each k , every unknown is a function of one variable:

conformal time τ , with $d\tau = dt/a(t)$. A PDE in space and time becomes a set of ODEs in τ , one set per k .

The ODEs



- **Background:** the Friedmann equation $\left(\frac{1}{a} \frac{da}{d\tau}\right)^2 = \frac{8\pi G}{3} a^2 \rho$, for the expansion $a(\tau)$.
- **Recombination:** an ODE for the free-electron fraction $x_e(\tau)$ as protons and electrons form hydrogen.
- **Perturbations, for each k :** density and velocity of dark matter and of baryons; the photon and neutrino distributions, expanded in multipoles ℓ ; and the metric potentials. For the photons,

$$\frac{dF_\ell}{d\tau} = \frac{k}{2\ell + 1} \left[\ell F_{\ell-1} - (\ell + 1) F_{\ell+1} \right] - a n_e \sigma_T F_\ell \quad (\ell \geq 3)$$

- **Stiff:** early on, the scattering rate $a n_e \sigma_T$ is enormous compared with the expansion rate. Codes use a tight-coupling approximation there, or an implicit solver.

Every pixel is an ODE

The Doubly Warped World of Binary Black Holes

NASA Goddard



Watch on

NASA Goddard, "The Doubly Warped World of Binary Black Holes": youtube.com/watch?v=rQcKIN9vj3U

References



- Further reading:
 - Qin et al., "Why is Boris algorithm so good?", *Phys. Plasmas* 20, 084503 (2013).
 - Berry & Healy, "Implementation of Gauss-Jackson integration for orbit propagation", *J. Astronaut. Sci.* 52, 331 (2004).
 - Frenkel & Smit, *Understanding Molecular Simulation*.
 - Ma & Bertschinger, *ApJ* 455, 7 (1995); Dodelson & Schmidt, *Modern Cosmology*.